Multi-Agent Financial Analysis System

AAI-520 Final Team Project — Brandon Wirgau, Christina, and Isaac

This notebook is the canonical, executable report for an agentic investment-research system. It accepts a stock ticker and user research question, uses LangGraph to route a bounded run to market, financial, news, or comprehensive specialists, evaluates and revises the complete report, and stores reusable lessons for later runs.

Repository: AAI-520 Final Project

> Research support only. The output is not personalized investment advice and does not guarantee future performance.

1. Project objectives and implemented capabilities

The project demonstrates four autonomous agent functions:

1. **Plan** research objectives, questions, and tool selection using prior-run memory as context.
2. **Use tools** to retrieve Yahoo Finance market/fundamental data, SEC EDGAR facts and filings, and multi-source company news.
3. **Reflect and evaluate** data quality, report grounding, and specialist-analysis completeness.
4. **Learn across runs** by persisting bounded, timestamped research lessons.

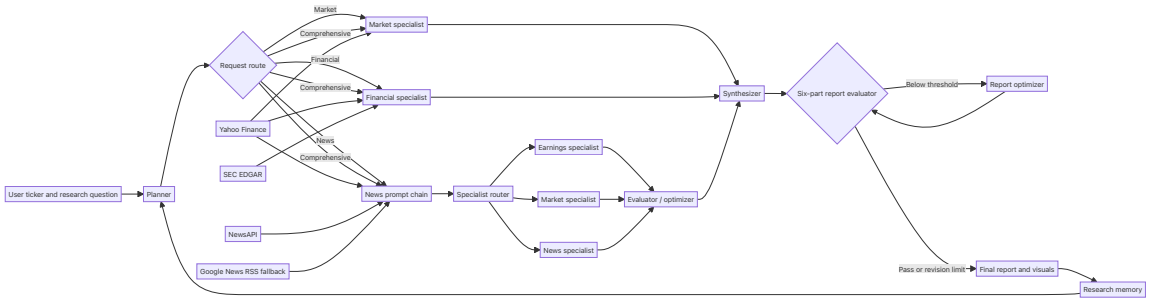
It also implements all three required workflow patterns:

- **Prompt chaining:** ingest → preprocess → classify → extract → summarize.
- **Routing:** the user request selects market-only, financial-only, news-only, or comprehensive research; news articles also route to earnings, market, and general-news specialists.
- **Evaluator–optimizer:** article analysis uses a four-part rubric, while the complete report is scored for completeness, grounding, reasoning, consistency, sources, and clarity and revised through a bounded LangGraph loop.

The refactored modules preserve the original notebook's planning, tool execution, validation, reflection, reporting, memory, SEC EDGAR retrieval, 10-year price graph, revenue and operating-expense analysis, multi-source news analysis, specialist routing, and iterative optimization. The updated Google News RSS fallback and retry-aware HTTP behavior are included as well.

2. Implemented system architecture

The research workflow keeps provider adapters, provider-independent tools, agents, workflows, memory, and presentation concerns separate. This makes each stage inspectable and independently testable.



3. Environment setup

Create and activate a virtual environment, install `requirements.txt`, start Ollama, and copy `.env.example` to `.env`. At minimum:

- `OLLAMA_MODEL=llama3.2` and a running `ollama serve` process;
- `SEC_USER_AGENT` containing an application name and monitored contact email;
- optional `NEWS_API_KEY` (Yahoo remains primary and Google RSS is a fallback);
- optional `DEFAULT_TICKER`, `RESEARCH_REQUEST`, `MAX_NEWS_ARTICLES`, `EVALUATION_SCORE_THRESHOLD`, `SUBSTANTIVE_REASONING_THRESHOLD`, `MAX_OPTIMIZER_ITERATIONS`, `REPORT_SCORE_THRESHOLD`, and `MAX_REPORT_REVISIONS`.

Credentials are read only from environment variables and are never displayed below.

In [1]:
```
import getpass
import html
import json
import os
import shutil
import subprocess
import sys
import tempfile
import time
import urllib.request
from pathlib import Path

try:
    import google.colab  # noqa: F401

    IN_COLAB = True
except ImportError:
    IN_COLAB = False
```

```
if IN_COLAB:
    repo_url = "https://github.com/bwirgau-usd/AAI-520-final-project-multi-a
    repo_dir = Path("/content/AAI-520-final-project-multi-agent-financial-an
    if not repo_dir.exists():
        subprocess.run(
            ["git", "clone", "--branch", "main", repo_url, str(repo_dir)],
            check=True,
        )
    os.chdir(repo_dir)
    subprocess.run(
        [sys.executable, "-m", "pip", "install", "-q", "-r", "requirements.t
        check=True,
    )

    if not os.getenv("NEWS_API_KEY"):
        news_key = getpass.getpass(
            "NEWS_API_KEY (optional; input hidden): "
        ).strip()
        if news_key:
            os.environ["NEWS_API_KEY"] = news_key
    if not os.getenv("SEC_USER_AGENT"):
        os.environ["SEC_USER_AGENT"] = input(
            "SEC user agent (name and contact email): "
        ).strip()

    if shutil.which("ollama") is None:
        subprocess.run(
            ["apt-get", "install", "-y", "-qq", "zstd"],
            check=True,
        )
        installer = urllib.request.urlopen(
            "https://ollama.com/install.sh",
            timeout=60,
        ).read()
        subprocess.run(["sh"], input=installer, check=True)
    subprocess.Popen(
        ["ollama", "serve"],
        stdout=subprocess.DEVNULL,
        stderr=subprocess.DEVNULL,
    )
    time.sleep(5)
    subprocess.run(
        ["ollama", "pull", os.getenv("OLLAMA_MODEL", "llama3.2")],
        check=True,
    )
    print("Colab environment ready.")
else:
    print("Local environment detected; start Ollama separately with ollama s
```

```
Local environment detected; start Ollama separately with ollama serve.
```

In [2]:

```
import os
import sys
from pathlib import Path
```

```
import pandas as pd
from IPython.display import Markdown, display
from dotenv import load_dotenv


def find_project_root(start: Path) -> Path:
    """Find the repository root from JupyterLab or a nested notebook path."""
    for candidate in (start.resolve(), *start.resolve().parents):
        if (candidate / "src").is_dir() and (candidate / "requirements.txt")
            return candidate
    raise FileNotFoundError("Could not locate the project root.")


PROJECT_ROOT = find_project_root(Path.cwd())
if str(PROJECT_ROOT) not in sys.path:
    sys.path.insert(0, str(PROJECT_ROOT))
load_dotenv(PROJECT_ROOT / ".env")

from src.agents import EvaluatorAgent, PlannerAgent, ResearchRequestRouter,
from src.data_sources.yahoo_finance import YahooFinanceClient
from src.memory import ResearchMemoryStore
from src.reporting import (
    plot_10_year_stock_price,
    plot_news_analysis_summary,
    plot_news_workflow,
    plot_operating_expenses,
    plot_revenue_and_operating_expenses,
    plot_system_architecture,
    render_console_summary,
)
from src.tools.registry import TOOLS
from src.workflows import build_research_workflow

TICKER = os.getenv("DEFAULT_TICKER", "AAPL").strip().upper()
RESEARCH_REQUEST = os.getenv(
    "RESEARCH_REQUEST",
    "How do market performance, financial condition, SEC evidence, and recen
).strip()
print(f"Project root: {PROJECT_ROOT}")
print(f"Ticker: {TICKER}")
print(f"Research request: {RESEARCH_REQUEST}")
print(f"NewsAPI configured: {bool(os.getenv('NEWS_API_KEY'))}")
print(f"SEC identity configured: {bool(os.getenv('SEC_USER_AGENT'))}")
```

```
/Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-gZAqjFTS/lib/pyt
hon3.13/site-packages/requests/__init__.py:113: RequestsDependencyWarning: u
rllib3 (2.7.0) or chardet (7.4.3)/charset_normalizer (3.4.4) doesn't match a
supported version!
  warnings.warn(
Project root: /Users/brandonwirgau/Projects/AAI/520/Final Project
Ticker: AAPL
Research request: How do market performance, financial condition, SEC eviden
ce, and recent news affect the company's current risk picture?
NewsAPI configured: False
SEC identity configured: True
```

```
/Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-gZAqjFTS/lib/pyt
hon3.13/site-packages/langgraph/cache/base/__init__.py:8: LangChainPendingDe
precationWarning: The default value of `allowed_objects` will change in a fu
ture version. Pass an explicit value (e.g., allowed_objects='messages' or al
lowed_objects='core') to suppress this warning.
  from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer
```

4. Agent design, data sources, and tool use

Component	Implemented responsibility
Planner Agent	Separates minimum required evidence from optional, reasoned tool choices and converts structured prior-run lessons into explicit objectives and success checks.
Tool Executor	Runs only registered tools and isolates provider errors so one failure does not terminate the research run.
Market Specialist / Tools	Retrieve price and volume history; compute one-month, three-month, and one-year returns, volatility, drawdown, 52-week range, and average volume.
Financial Specialist / Tools	Retrieve income, cash-flow, and balance-sheet statements; preserve or derive operating expenses; compute revenue/earnings growth; and analyze cash, debt, net debt, assets, liabilities, and equity.
SEC EDGAR Tool	Resolves CIK, reads official company facts, and lists recent 10-K, 10-Q, and 8-K filings with source metadata.
News Prompt Chain	Combines Yahoo and NewsAPI; invokes Google RSS when primary coverage is missing or sparse; deduplicates and structures articles.
Specialist Router	Sends classified items to earnings, market, or general-news specialists.
Evaluator / Optimizer	Scores structural completeness separately from substantive financial reasoning; either failing score triggers bounded revision and unresolved work retains an insufficient-evidence status.
Synthesis Agent	Produces a grounded Markdown report spanning market, financial, filing, and news evidence.
Memory Curator / Store	Saves each weakness with a concrete next-run action and success check using atomic file replacement; memory is planning context, never current evidence.

FRED and Alpha Vantage packages remain deliberate extension points. The report does not claim they are integrated data sources.

In [3]:
```
tool_rows = [
    {
        "tool": name,
        "role": {
            "price_data": "1m/3m/1y performance, risk, range, and volume",
            "company_info": "Profile, valuation, margins, and dividend",
            "financials": "Revenue, operating-expense, profitability, and EP
            "cash_flow": "Operating cash flow, free cash flow, and capex",
```

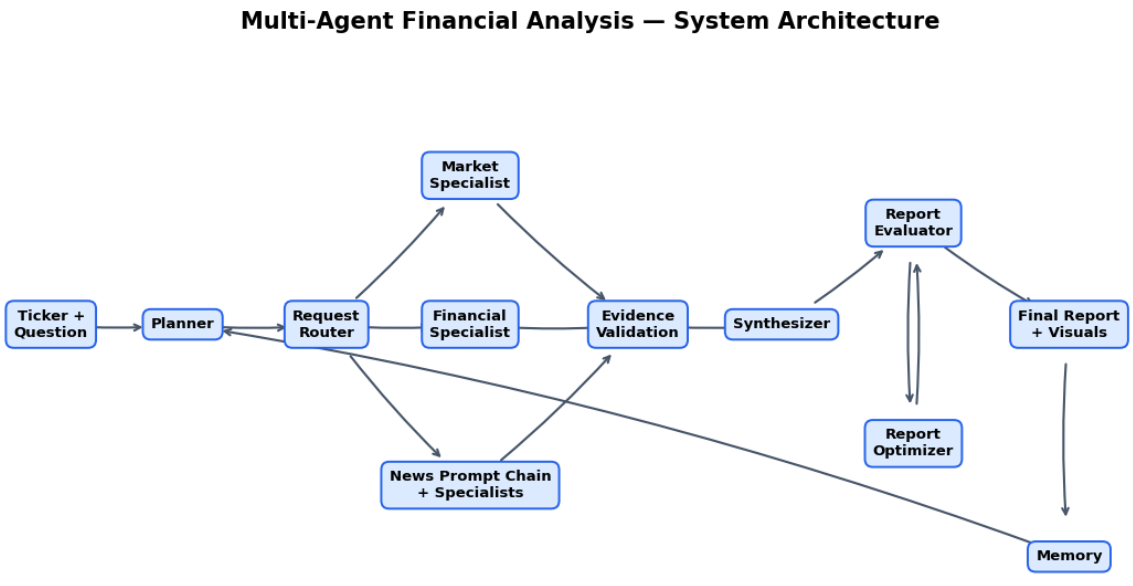
```
            "balance_sheet": "Cash, debt, assets, liabilities, and equity",
            "financial_growth": "Revenue, net-income, and EPS growth",
            "sec_edgar": "Official facts and recent filings",
        }.get(name, "Research evidence"),
    }
    for name in TOOLS
]
display(pd.DataFrame(tool_rows))
```

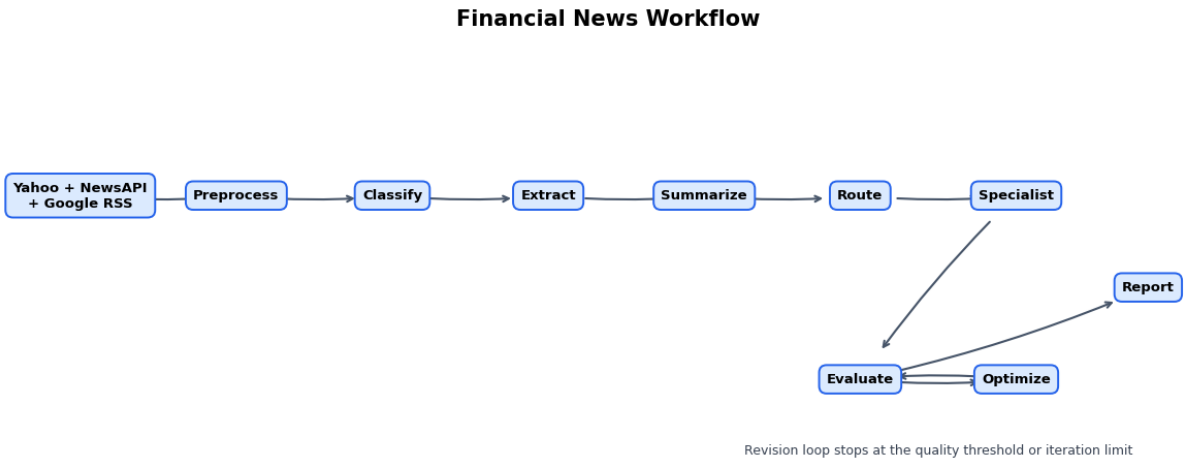
	tool	role
0	price_data	1m/3m/1y performance, risk, range, and volume
1	company_info	Profile, valuation, margins, and dividend
2	financials	Revenue, operating-expense, profitability, and...
3	cash_flow	Operating cash flow, free cash flow, and capex
4	balance_sheet	Cash, debt, assets, liabilities, and equity
5	financial_growth	Revenue, net-income, and EPS growth
6	sec_edgar	Official facts and recent filings

5. Workflow diagrams

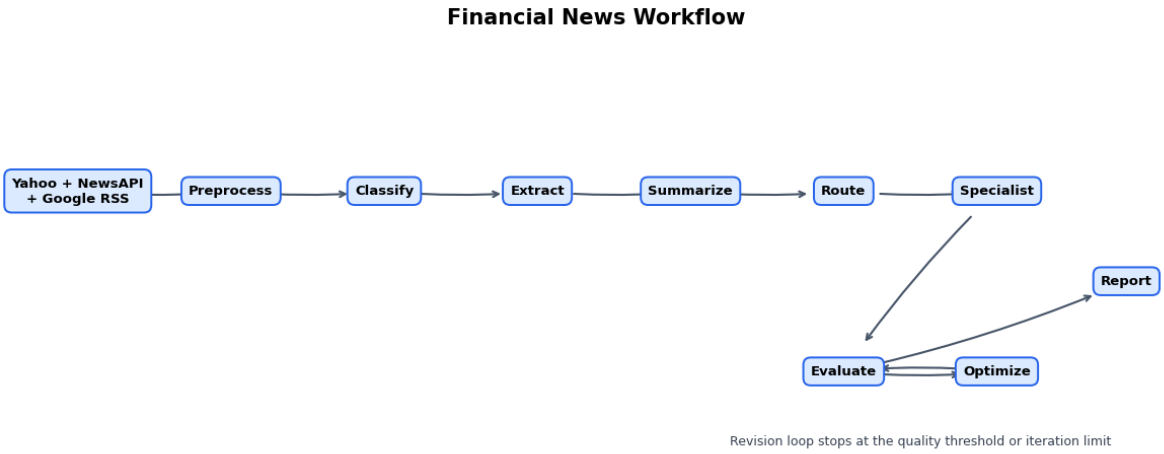
These diagrams are generated by the project's reporting module, so the submitted report reflects the implemented code rather than a disconnected illustration.

In [4]:
```
plot_system_architecture()
plot_news_workflow()
```



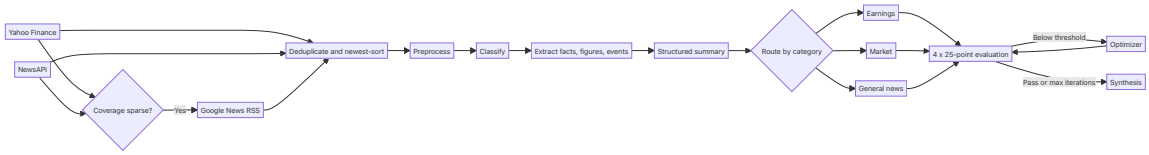


Out[4]:



6. Prompt chaining, routing, and evaluator–optimizer logic

The updated news workflow uses provider isolation and explicit fallback logic:



Provider failures are retained in `source_errors`; original and optimized analyses plus every evaluation score remain in workflow state for auditability. NewsAPI and RSS use retry-enabled HTTP sessions for transient 429/5xx and connection failures.

7. End-to-end investment research run

The next cell performs two LangGraph runs for the same request. Run 1 collects evidence and saves evaluator feedback. Run 2 receives the structured lesson and reanalyzes the exact Run 1 tool observations and source articles, so any quality change cannot be attributed to different provider responses. Each run:

1. accepts and classifies the research request;
2. conditionally routes to market, financial, news, or all specialists;
3. retrieves Yahoo Finance, SEC, and prompt-processed news evidence as selected;
4. validate observations;
5. reflect on evidence quality;
6. synthesize the report;
7. scores the complete report across six criteria;
8. revises and re-evaluates below-threshold reports up to a configured limit; and
9. saves reusable evaluator feedback when memory is enabled.

A running local Ollama model and network access are required for a complete live result. Missing optional providers are recorded rather than treated as fatal.

In [5]:
```
demo_memory_dir = Path(tempfile.mkdtemp(prefix="aai520-notebook-memory-"))
demo_memory_store = ResearchMemoryStore(demo_memory_dir / "research_memory.j
workflow = build_research_workflow(
    project_root=PROJECT_ROOT,
    memory_store=demo_memory_store,
)
print("FIRST RUN — establish evidence and save feedback")
first_result = workflow.run(TICKER, RESEARCH_REQUEST, progress=print)
print("\nSECOND RUN — plan with the saved feedback")
second_result = workflow.run(
    TICKER,
    RESEARCH_REQUEST,
    remember=False,
    progress=print,
    evidence_snapshot=first_result,
)
result = second_result
```

```
FIRST RUN — establish evidence and save feedback
Planning and routing the research request
Running all selected research specialists
Running market tool: price_data
Running financial tool: sec_edgar
Running financial tool: financials
Processing news article 1/8
Processing news article 2/8
Processing news article 3/8
Processing news article 4/8
Processing news article 5/8
Processing news article 6/8
Processing news article 7/8
Processing news article 8/8
Validating specialist observations
Reflecting on evidence quality
Synthesizing the draft report
Scoring the complete report
Revising the complete report (iteration 1)
Scoring the complete report
Finalizing the evaluated report
Saving evaluator feedback for a future run

SECOND RUN — plan with the saved feedback
Planning and routing the research request
Running all selected research specialists
Processing news article 1/8
Processing news article 2/8
Processing news article 3/8
Processing news article 4/8
Processing news article 5/8
Processing news article 6/8
Processing news article 7/8
Processing news article 8/8
Validating specialist observations
Reflecting on evidence quality
Synthesizing the draft report
Scoring the complete report
Revising the complete report (iteration 1)
Scoring the complete report
Finalizing the evaluated report
```

7.1 Structured planning and quality artifacts

In [6]:
```python
review_artifacts = {
    "symbol": result["symbol"],
    "research_request": result.get("research_request"),
    "request_route": result.get("request_route"),
    "selected_specialists": result.get("selected_specialists", []),
    "plan": result.get("plan", {}),
    "specialist_analyses": result.get("specialist_analyses", {}),
    "validation": result.get("validation", []),
    "reflection": result.get("reflection", {}),
    "report_evaluation": result.get("report_evaluation", {}),
    "report_revision_history": result.get("report_revision_history", []),
```

```
    "graph_trace": result.get("graph_trace", []),
}
print(json.dumps(review_artifacts, indent=2, default=str))
display(pd.DataFrame([{
    "research_request": result.get("research_request"),
    "selected_route": result.get("request_route"),
    "selected_specialists": ", ".join(result.get("selected_specialists", []))
}]))
```

```
{
  "symbol": "AAPL",
  "research_request": "How do market performance, financial condition, SEC e
vidence, and recent news affect the company's current risk picture?",
  "request_route": "comprehensive",
  "selected_specialists": [
    "market",
    "financial",
    "news"
  ],
  "plan": {
    "objectives": [
      "Market performance",
      "Financial condition",
      "SEC evidence",
      "Recent news",
      "Apply prior-run feedback: Explain how each material event could affec
t revenue, costs, cash flow, or valuation",
      "Apply prior-run feedback: Address this evaluator finding explicitly:
Financial transmission mechanism",
      "Apply prior-run feedback: Address this evaluator finding explicitly:
Specific financial figures or details on the impact of events"
    ],
    "tools": [
      "price_data",
      "sec_edgar",
      "financials"
    ],
    "required_tools": [
      "price_data",
      "sec_edgar",
      "financials"
    ],
    "optional_tools": [],
    "available_optional_tools": [
      "balance_sheet",
      "cash_flow",
      "company_info",
      "financial_growth"
    ],
    "questions": [
      "How has market performance affected AAPL's current risk picture?",
      "What is AAPL's current financial condition?",
      "What SEC evidence is available for AAPL?",
      "How have recent news events impacted AAPL's risk profile?",
      "How does this run satisfy the saved success check: Include supporting
article evidence, a financial implication, and uncertainty; state explicitly
when no material effect is established?",
      "How does this run satisfy the saved success check: The next report su
pplies relevant evidence and states remaining uncertainty for this finding?"
    ],
    "research_request": "How do market performance, financial condition, SEC
evidence, and recent news affect the company's current risk picture?",
    "route": "comprehensive",
    "specialists": [
      "market",
```

```
      "financial",
      "news"
    ],
    "memory_context": [
      {
        "date": "2026-10-18T03:10:02.449603+00:00",
        "symbol": "AAPL",
        "summary": "Here's a summary of reusable research lessons from the A
APL run in 120 words:\n\nEvidence coverage is essential for a comprehensive
risk picture. However, data quality issues arise due to missing financial fi
gures and uncertain event significance. To improve, researchers should verif
y financial transmission mechanisms and cite supporting article evidence. Cl
arity and consistency are also crucial for a well-structured report. The ver
ification scope should explicitly compare filing periods and provider statem
ent periods. The use of deterministic observations for verified financial me
trics can help. The interpretation should consider both historical return an
d measured price risk. Finally, researchers should verify detected events an
d acknowledge uncertainty in financial significance. These lessons can infor
m future research and improve the quality of AAPL investment research report
s.",
        "weaknesses": [
          "Substantive financial reasoning and event significance"
        ],
        "lessons": [
          {
            "weakness": "News analysis lacks financial implications",
            "next_run_action": "Explain how each material event could affect
revenue, costs, cash flow, or valuation",
            "success_check": "Include supporting article evidence, a financi
al implication, and uncertainty; state explicitly when no material effect is
established"
          },
          {
            "weakness": "Substantive financial reasoning and event significa
nce",
            "next_run_action": "Explain how each material event could affect
revenue, costs, cash flow, or valuation",
            "success_check": "Include supporting article evidence, a financi
al implication, and uncertainty; state explicitly when no material effect is
established"
          },
          {
            "weakness": "Financial transmission mechanism",
            "next_run_action": "Address this evaluator finding explicitly: F
inancial transmission mechanism",
            "success_check": "The next report supplies relevant evidence and
states remaining uncertainty for this finding"
          },
          {
            "weakness": "Specific financial figures or details on the impact
of events",
            "next_run_action": "Address this evaluator finding explicitly: S
pecific financial figures or details on the impact of events",
            "success_check": "The next report supplies relevant evidence and
states remaining uncertainty for this finding"
          }
```

```
        ]
      }
    ],
    "applied_memory_actions": [
      {
        "weakness": "News analysis lacks financial implications",
        "next_run_action": "Explain how each material event could affect rev
enue, costs, cash flow, or valuation",
        "success_check": "Include supporting article evidence, a financial i
mplication, and uncertainty; state explicitly when no material effect is est
ablished"
      },
      {
        "weakness": "Financial transmission mechanism",
        "next_run_action": "Address this evaluator finding explicitly: Finan
cial transmission mechanism",
        "success_check": "The next report supplies relevant evidence and sta
tes remaining uncertainty for this finding"
      },
      {
        "weakness": "Specific financial figures or details on the impact of
events",
        "next_run_action": "Address this evaluator finding explicitly: Speci
fic financial figures or details on the impact of events",
        "success_check": "The next report supplies relevant evidence and sta
tes remaining uncertainty for this finding"
      }
    ]
  },
  "specialist_analyses": {
    "market": {
      "performance": {
        "1m_return_percent": 1.92,
        "3m_return_percent": 5.26,
        "1y_return_percent": 36.64
      },
      "risk": {
        "annualized_volatility_percent": 24.4,
        "maximum_drawdown_percent": -13.8
      },
      "range_and_volume": {
        "52_week_high": 348.15,
        "52_week_low": 242.76,
        "latest_volume": 59895300.0,
        "average_volume": 48103449.4
      }
    },
    "financial": {
      "operating_expenses": {
        "2025-09-30 00:00:00": 62151000000.0,
        "2024-09-30 00:00:00": 57467000000.0,
        "2023-09-30 00:00:00": 54847000000.0,
        "2022-09-30 00:00:00": 51345000000.0
      },
      "growth": {},
      "cash_and_debt": {},
```

```
      "balance_sheet_trends": {},
      "has_sec_evidence": true
    },
    "news": {
      "article_count": 8,
      "average_score": 100.0,
      "average_structural_score": 100.0,
      "average_substantive_reasoning_score": 50.0,
      "routing_counts": {
        "market": 7,
        "earnings": 1
      }
    }
  },
  "validation": [],
  "reflection": {
    "strengths": [
      "Structural completeness passed"
    ],
    "weaknesses": [
      "Substantive financial reasoning and event significance"
    ],
    "missing_information": [
      "Financial transmission mechanism",
      "Specific financial figures or details on the impact of events"
    ],
    "suspicious_values": [],
    "follow_up_questions": [
      "Explain how each material event could affect revenue, costs, cash flo
w, or valuation",
      "Address this evaluator finding explicitly: Financial transmission mec
hanism",
      "Address this evaluator finding explicitly: Specific financial figures
or details on the impact of events"
    ],
    "deterministic_issues": []
  },
  "report_evaluation": {
    "score": 89.17,
    "passed": true,
    "threshold": 75.0,
    "minimum_criterion_score": 60.0,
    "criteria": {
      "completeness": 100.0,
      "grounding": 100.0,
      "reasoning": 75.0,
      "consistency": 100.0,
      "sources": 60.0,
      "clarity": 100.0
    },
    "feedback": [
      "Improve sources; current score is 60/100."
    ],
    "validation_issues": []
  },
  "report_revision_history": [
```

```
    {
      "iteration": 1,
      "before_score": 81.67,
      "after_score": 89.17,
      "feedback": [
        "Improve reasoning; current score is 30/100.",
        "Improve sources; current score is 60/100.",
        "No report can pass while these criteria are below 60/100: reasonin
g."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```


	research_request	selected_route	selected_specialists
0	How do market performance, financial condition...	comprehensive	market, financial, news

7.2 Collected market, fundamental, and SEC evidence

The main report shows compact evidence tables so the important values and reporting periods remain reviewable. The complete observation object is retained in the companion audit HTML generated in the appendix. Provider errors appear alongside successful results instead of being hidden.

In [7]:
```
observations = result.get("observations", {})
market_fields = {
    "price_data": ["latest_price", "1m_return_percent", "3m_return_percent",
    "company_info": ["longName", "sector", "industry", "marketCap", "enterpr
}
snapshot_rows = []
for source_name, field_names in market_fields.items():
    payload = observations.get(source_name, {})
    for field_name in field_names:
        if field_name in payload:
            snapshot_rows.append({"source": source_name, "metric": field_nam
display(Markdown("#### Market and company snapshot"))
display(pd.DataFrame(snapshot_rows))

statement_rows = []
for source_name in ("financials", "cash_flow"):
    for metric, periods in observations.get(source_name, {}).items():
```

```
        if isinstance(periods, dict) and periods:
            period, value = next(iter(periods.items()))
            statement_rows.append({"source": source_name, "metric": metric,
display(Markdown("#### Latest income-statement and cash-flow observations"))
display(pd.DataFrame(statement_rows))

balance_sheet = observations.get("balance_sheet", {})
balance_period = balance_sheet.get("trends", {}).get("latest_period")
balance_rows = [{"metric": key, "period": balance_period, "value": value} fo
growth_rows = [{"metric": key, "period": f"{observations.get('financial_grow
display(Markdown("#### Balance sheet and deterministic growth"))
display(pd.DataFrame(balance_rows + growth_rows))

sec_evidence = observations.get("sec_edgar", {})
sec_fact_rows = [{"fact": label, **fact} for label, fact in sec_evidence.get
display(Markdown("#### SEC company facts and filing periods"))
display(pd.DataFrame(sec_fact_rows))
display(pd.DataFrame(sec_evidence.get("recent_filings", [])))
if sec_evidence.get("error"):
    print(f"SEC EDGAR: {sec_evidence['error']}")
print("Complete raw observations are preserved in the companion audit HTML."
```

Market and company snapshot

	source	metric	value
0	price_data	latest_price	343.46
1	price_data	1m_return_percent	1.92
2	price_data	3m_return_percent	5.26
3	price_data	1y_return_percent	36.64
4	price_data	annualized_volatility_percent	24.40
5	price_data	maximum_drawdown_percent	-13.80
6	price_data	52_week_high	348.15
7	price_data	52_week_low	242.76
8	price_data	latest_volume	59895300.00
9	price_data	average_volume	48103449.40

Latest income-statement and cash-flow observations

	source	metric	period	value
0	financials	Total Revenue	2025-09-30 00:00:00	4.161610e+11
1	financials	Operating Expenses	2025-09-30 00:00:00	6.215100e+10
2	financials	Operating Income	2025-09-30 00:00:00	1.330500e+11
3	financials	Net Income	2025-09-30 00:00:00	1.120100e+11
4	financials	EBITDA	2025-09-30 00:00:00	1.447480e+11
5	financials	Diluted EPS	2025-09-30 00:00:00	7.460000e+00

Balance sheet and deterministic growth

—

SEC company facts and filing periods

	fact	value	unit	fiscal_year	fiscal_period	form	period_end	
0	Revenue	3.643570e+11	USD	2026	Q3	10-Q	2026-06-27	20
1	Net Income	1.014640e+11	USD	2026	Q3	10-Q	2026-06-27	20
2	Operating Income	1.224320e+11	USD	2026	Q3	10-Q	2026-06-27	20
3	Operating Expenses	5.635000e+10	USD	2026	Q3	10-Q	2026-06-27	20
4	Operating Cash Flow	1.169960e+11	USD	2026	Q3	10-Q	2026-06-27	20
5	Assets	3.832660e+11	USD	2026	Q3	10-Q	2026-06-27	20
6	Liabilities	2.757460e+11	USD	2026	Q3	10-Q	2026-06-27	20
7	Stockholders Equity	1.075200e+11	USD	2026	Q3	10-Q	2026-06-27	20
8	Diluted EPS	6.880000e+00	USD/ shares	2026	Q3	10-Q	2026-06-27	20

	form	filing_date	report_date	accession	filing_url
0	10-Q	2026-07-31	2026-06-27	0000320193-26-000020	https://www.sec.gov/Archives/edgar/data/320193...
1	8-K	2026-07-30	2026-07-30	0000320193-26-000018	https://www.sec.gov/Archives/edgar/data/320193...
2	10-Q	2026-05-01	2026-03-28	0000320193-26-000013	https://www.sec.gov/Archives/edgar/data/320193...
3	8-K	2026-04-30	2026-04-30	0000320193-26-000011	https://www.sec.gov/Archives/edgar/data/320193...
4	8-K	2026-04-20	2026-04-17	0001140361-26-015711	https://www.sec.gov/Archives/edgar/data/320193...
5	8-K	2026-02-24	2026-02-24	0001140361-26-006577	https://www.sec.gov/Archives/edgar/data/320193...
6	10-Q	2026-01-30	2025-12-27	0000320193-26-000006	https://www.sec.gov/Archives/edgar/data/320193...
7	8-K	2026-01-29	2026-01-29	0000320193-26-000005	https://www.sec.gov/Archives/edgar/data/320193...

Complete raw observations are preserved in the companion audit HTML.

7.3 Detailed news analysis

This table surfaces source provenance, routing decisions, final evaluator scores, optimizer activity, and the final acceptance status. The structural score measures field completeness and source traceability; the separate substantive score measures financial reasoning and event significance. Either failed threshold triggers revision. A 100 structural score must not be interpreted as perfect financial analysis. Complete per-article records remain in the companion audit HTML.

In [8]:
```
news_analysis = result.get("news_analysis", {})
news_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    evaluation = article.get("evaluation", {})
    news_rows.append({
        "article": index,
        "source": article.get("source"),
        "published_at": article.get("published_at"),
        "headline": article.get("title"),
        "category": article.get("category"),
        "classification_confidence": article.get("classification_confidence"
```

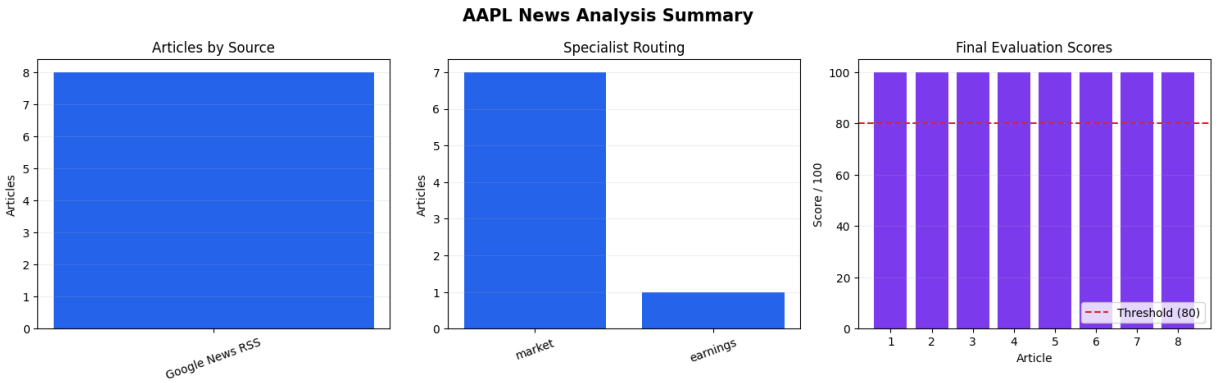
```
        "company": article.get("extracted", {}).get("company"),
        "event": article.get("extracted", {}).get("event"),
        "event_dates": article.get("extracted", {}).get("event_dates"),
        "sentiment": article.get("extracted", {}).get("sentiment"),
        "specialist": article.get("specialist"),
        "structural_score": evaluation.get("score"),
        "structural_passed": evaluation.get("passed"),
        "substantive_reasoning_score": evaluation.get("substantive_reasoning
        "substantive_reasoning_passed": evaluation.get("substantive_reasonin
        "overall_passed": evaluation.get("overall_passed"),
        "analysis_status": article.get("analysis_status"),
        "optimizer_revisions": len(article.get("optimization_history", [])),
        "url": article.get("url"),
    })

display(pd.DataFrame(news_rows))
print(json.dumps({
    "pipeline_stages": news_analysis.get("pipeline_stages", []),
    "retrieved_at": news_analysis.get("retrieved_at"),
    "retrieved_counts": news_analysis.get("retrieved_counts", {}),
    "source_errors": news_analysis.get("source_errors", {}),
    "source_counts": news_analysis.get("source_counts", {}),
    "routing_counts": news_analysis.get("routing_counts", {}),
    "average_structural_score": news_analysis.get("average_structural_score"
    "average_substantive_reasoning_score": news_analysis.get("average_substa
    "score_scope": news_analysis.get("score_scope"),
    "passed_count": news_analysis.get("passed_count", 0),
    "structural_passed_count": news_analysis.get("structural_passed_count",
    "substantive_reasoning_threshold": news_analysis.get("substantive_reason
    "evidence_reused": news_analysis.get("evidence_reused"),
}, indent=2, default=str))
```

	article	source	published_at	headline	category	classification_conf
0	1	Google News RSS	2026-10-17T14:50:51+00:00	Apple Inc. $AAPL Stock: HWG Holdings LP Raises...	market	
1	2	Google News RSS	2026-10-17T14:50:51+00:00	3,459 Apple Inc. $AAPL Shares Bought by Titlei...	market	
2	3	Google News RSS	2026-10-17T13:39:16+00:00	Apple Inc. (AAPL) Stock Forecasts - Yahoo! Fin...	market	
3	4	Google News RSS	2026-10-17T07:37:28+00:00	Apple Inc. $AAPL Stock Sold by Chickasaw Capit...	market	
4	5	Google News RSS	2026-10-17T07:37:28+00:00	30,425 Apple Inc. $AAPL Shares Sold by Everhar...	market	
5	6	Google News RSS	2026-10-17T07:28:28+00:00	Apple Inc. (AAPL) Price: Live Chart & Market C...	market	
6	7	Google News RSS	2026-10-17T06:33:25+00:00	Apple Inc. (AAPL) - What Are Researchers Sayin...	earnings	
7	8	Google News RSS	2026-10-13T12:00:02+00:00	Apple Inc. (AAPL) is Attracting Investor Atten...	market	

```
{
  "pipeline_stages": [
    "ingest",
    "preprocess",
    "classify",
    "extract",
    "summarize",
    "route",
    "specialist",
    "evaluate",
    "optimize"
  ],
  "retrieved_at": "2026-10-18T03:11:14.861722+00:00",
  "retrieved_counts": {
    "evidence_replay": 8
  },
  "source_errors": {},
  "source_counts": {
    "Google News RSS": 8
  },
  "routing_counts": {
    "market": 7,
    "earnings": 1
  },
  "average_structural_score": 100.0,
  "average_substantive_reasoning_score": 50.0,
  "score_scope": "Structural completeness and source traceability are scored
separately from substantive financial reasoning and event significance.",
  "passed_count": 2,
  "structural_passed_count": 8,
  "substantive_reasoning_threshold": 60.0,
  "evidence_reused": true
}
```


In [9]:
```
news_figure = plot_news_analysis_summary(news_analysis)
if news_figure is None:
    display(Markdown("*No news-summary chart is available because no article
```



8. Final research report

The model-generated report uses only collected observations and explicitly distinguishes evidence, interpretation, risk, uncertainty, and missing data.

In [10]:
```
display(Markdown(result["report"]))
```

Company Overview

- Company Name: Apple Inc.
- Symbol: AAPL
- Sector: Technology
- Industry: Electronics
- Country: United States
- Current Price: $343.46
- Market Capitalization: $2.45 Trillion
- Enterprise Value: $1.15 Trillion

Price Performance

- 1-Year Return: 36.64%
- 1-Month Return: 1.92%
- 3-Month Return: 5.26%

Valuation and Profitability

- Trailing P/E: 23.55
- Forward P/E: 23.65
- Price/Sales: 10.33
- Profit Margin: 24.1%
- Operating Margin: 24.4%
- Return on Equity: 34.5%
- Beta: 1.23

Financial Performance

Cash Flow

- Operating Cash Flow: $116,996,000,000
- Free Cash Flow: $114,510,000,000
- Capital Expenditure: $13,439,000,000
- Dividend Yield: 1.37%

SEC EDGAR Evidence

- SEC Entity Name: Apple Inc.
- CIK: 0000320193
- SIC Classification: 3571
- Official Company Facts:
  - Operating Cash Flow (2026 Q3): $116,996,000,000
  - Assets (2026 Q3): $383,266,000,000
  - Liabilities (2026 Q3): $275,746,000,000
  - Stockholders Equity (2026 Q3): $107,520,000,000

Dividend

- Dividend Yield: 1.37%

Recent News and Catalysts

- News Source: MarketBeat
- Publisher: MarketBeat
- Publication Date: 2026-10-17
- Headline: "Apple Inc. $AAPL Stock: HWG Holdings LP Raises Stake - MarketBeat"
- URL: https://news.google.com/rss/articles/CBMirwFBVV95cUxNb3MyM19UNlZUZDRYVEN0azN1QzA4bjRUVFJzejZUVGVWb0w1MVl~oc=5"
- Detected Events: Stake Raise
- Figures: $1,000,000,000
- Routed Specialist: Market
- Structural Completeness/Traceability Score: 100
- Feedback: Structural completeness passed, but substantive reasoning remains insufficient.
- Interpretation: The article provides evidence relevant to share price and market activity.
- Uncertainties: The provider payload may contain only a headline or snippet; verify the linked source.

Risks and Uncertainties

- Risks: Market risk, industry risk
- Uncertainties: Provider payload may contain only a headline or snippet; verify the linked source.

Data Quality

- Missing Information: Financial transmission mechanism, specific financial figures or details on the impact of events.
- Suspicious Values: None.

Further Research

- Conduct a more in-depth analysis of the article's financial implications and verify the financial transmission mechanism.

Verification Scope

SEC facts provide primary-source evidence for selected financial metrics. Market data and news remain provider-supplied and require independent verification.

Verified Financial Metrics

The following values are rendered deterministically from provider observations; no model arithmetic is used.

- **Start Price**: $251.36
- **Latest Price**: $343.46
- **One-Month Return**: 1.92%
- **Three-Month Return**: 5.26%
- **One-Year Return**: 36.64%
- **Annualized Volatility**: 24.40%
- **Maximum Drawdown**: -13.80%
- **52-Week High**: $348.15
- **52-Week Low**: $242.76
- **Latest Volume**: 59,895,300
- **Average Volume**: 48,103,449
- **Price Observation Count**: 251
- **Revenue** (2025-09-30 00:00:00): $416,161,000,000
- **Operating Expenses** (2025-09-30 00:00:00): $62,151,000,000
- **Operating Income** (2025-09-30 00:00:00): $133,050,000,000
- **Net Income** (2025-09-30 00:00:00): $112,010,000,000
- **EBITDA** (2025-09-30 00:00:00): $144,748,000,000
- **Diluted EPS** (2025-09-30 00:00:00): $7.46

- **EBITDA / Net Income** (2025-09-30 00:00:00): 1.292x

Evidence-Based Interpretation

Because the one-year return was 36.64% while annualized volatility was 24.40% and maximum drawdown was -13.80%, the historical return must be considered together with measured price risk; these observations do not predict future performance.

SEC EDGAR supplied 9 official company fact(s) and 8 recent filing record(s), providing primary-source corroboration; however, filing periods and provider statement periods should be compared explicitly before drawing conclusions.

The news workflow processed 8 article(s). The structural completeness and source-traceability rubric averaged 100.00%. The separate substantive financial-reasoning and event-significance score averaged 50.00%. A 100% structural score means the expected fields and traceable source evidence were present; it does not establish perfect financial analysis. Because provider payloads may contain only headlines or snippets, detected events remain uncertain and should be verified at the linked publishers.

Taken together, the market measurements, financial condition, SEC evidence, and current news indicate a mixed, evidence-bounded risk picture rather than a forecast or an investment recommendation.

Applied Prior-Run Feedback

- **Saved weakness:** News analysis lacks financial implications
  - **Action applied:** Explain how each material event could affect revenue, costs, cash flow, or valuation
  - **Success check:** Include supporting article evidence, a financial implication, and uncertainty; state explicitly when no material effect is established
  - **Execution evidence:** The news specialist applied the action to 8 article(s); 2 passed the substantive-reasoning check. Representative interpretation: The article provides evidence of market activity and stake increase, but lacks specific financial details to assess the impact on revenue, costs, cash flow, or valuation.
- **Saved weakness:** Financial transmission mechanism
  - **Action applied:** Address this evaluator finding explicitly: Financial transmission mechanism
  - **Success check:** The next report supplies relevant evidence and states remaining uncertainty for this finding
  - **Execution evidence:** The planner and synthesizer received this action. Its success check requires reviewer verification against the report evidence.
- **Saved weakness:** Specific financial figures or details on the impact of events
  - **Action applied:** Address this evaluator finding explicitly: Specific financial figures

or details on the impact of events

- **Success check:** The next report supplies relevant evidence and states remaining uncertainty for this finding
- **Execution evidence:** The planner and synthesizer received this action. Its success check requires reviewer verification against the report evidence.

8.1 Deterministic report view

This second rendering guarantees that the core company overview, price performance, valuation, financial performance, cash flow, dividend, SEC filing evidence, news-source coverage, validation findings, and missing information remain visible even if model prose varies between runs.

In [11]:
```
print(render_console_summary(result, include_article_details=False))
```

```
================================================================================
FINAL RESEARCH REPORT
================================================================================

AAPL (AAPL)
--------------------------------------------------------------------------------

1. COMPANY OVERVIEW
   Sector:              N/A
   Industry:            N/A
   Country:             N/A
   Current Price:       $343.46
   Market Cap:          N/A
   Enterprise Value:    N/A

2. PRICE PERFORMANCE
   Start Price:         $251.36
   Latest Price:        $343.46
   One-Year Return:     36.64%
   Annualized Volatility: 24.40%
   Maximum Drawdown:    -13.80%
   Observations:        251

3. VALUATION
   Trailing P/E:        N/A
   Forward P/E:         N/A
   Price/Sales (TTM):   N/A
   Profit Margin:       N/A
   Operating Margin:    N/A
   Return on Equity:    N/A
   Beta:                N/A

4. FINANCIAL PERFORMANCE (2025)
   Revenue:             $416.161B
   Operating Expenses:  $62.151B
   Operating Income:    $133.050B
   Net Income:          $112.010B
   EBITDA:              $144.748B
   Diluted EPS:         $7.46
   EBITDA / Net Income: 1.292x

5. CASH FLOW (2025)
   Operating Cash Flow: N/A
   Free Cash Flow:      N/A
   Capital Expenditure: N/A

6. BALANCE SHEET AND GROWTH
   Cash:                N/A
   Total Debt:          N/A
   Net Debt:            N/A
   Cash / Debt:         N/A
   Total Assets:        N/A
   Total Liabilities:   N/A
   Stockholders' Equity: N/A
   Revenue Growth:      N/A
   Net Income Growth:   N/A
```

Diluted EPS Growth: N/A
Cash Trend: N/A
Debt Trend: N/A
Assets Trend: N/A

7. SEC EDGAR EVIDENCE
Source: SEC EDGAR
Entity: Apple Inc.
CIK: 0000320193
SIC: 3571 — Electronic Computers
Official Company Facts:
Revenue: 364,357,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Net Income: 101,464,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Operating Income: 122,432,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Operating Expenses: 56,350,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Operating Cash Flow: 116,996,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Assets: 383,266,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Liabilities: 275,746,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Stockholders Equity: 107,520,000,000.00 USD (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Diluted EPS: 6.88 USD/shares (FY 2026; period Q3; ended 2026-06-27; filed 2026-07-31; form 10-Q; accession 0000320193-26-000020)
Recent Filings:
- 10-Q | filed 2026-07-31 | report 2026-06-27 | accession 0000320193-26-000020 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm
- 8-K | filed 2026-07-30 | report 2026-07-30 | accession 0000320193-26-000018 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000018/aapl-20260730.htm
- 10-Q | filed 2026-05-01 | report 2026-03-28 | accession 0000320193-26-000013 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000013/aapl-20260328.htm
- 8-K | filed 2026-04-30 | report 2026-04-30 | accession 0000320193-26-000011 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000011/aapl-20260430.htm
- 8-K | filed 2026-04-20 | report 2026-04-17 | accession 0001140361-26-015711 | https://www.sec.gov/Archives/edgar/data/320193/000114036126015711/ef20071035_8k.htm
- 8-K | filed 2026-02-24 | report 2026-02-24 | accession 0001140361-26-006577 | https://www.sec.gov/Archives/edgar/data/320193/000114036126006577/ef20060722_8k.htm
- 10-Q | filed 2026-01-30 | report 2025-12-27 | accession 0000320193-26-000006 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000006/aapl-20251227.htm
- 8-K | filed 2026-01-29 | report 2026-01-29 | accession 0000320193-26-000005 | https://www.sec.gov/Archives/edgar/data/320193/000032019326000005/aapl-20260129.htm

8. DIVIDEND

```
   Dividend Yield:      N/A

9. RECENT NEWS AND CATALYSTS
   Retrieved At:        2026-10-18T03:11:14.861722+00:00
   Pipeline Stages:     ingest → preprocess → classify → extract → summarize
→ route → specialist → evaluate → optimize
   Articles Processed:  8
   Retrieved Counts:    evidence_replay=8
   Source Counts:       Google News RSS=8
   Routing Counts:      market=7, earnings=1
   Avg Structural Score: 100.0/100
   Avg Substantive Score: 50.0/100
   Passed Evaluations:  2
   Score Threshold:     80.0/100
   Optimizer Limit:     3
   Score Meaning:       Structural scores measure completeness and traceabil
ity, not perfect financial analysis.
   Article Details:    Saved in the companion audit HTML artifact.

10. RISKS AND UNCERTAINTIES
   - Substantive financial reasoning and event significance

11. DATA QUALITY
   - Evidence strength: Structural completeness passed

12. MISSING INFORMATION
   - Financial transmission mechanism
   - Specific financial figures or details on the impact of events

13. FURTHER RESEARCH
   - Explain how each material event could affect revenue, costs, cash flow,
or valuation
   - Address this evaluator finding explicitly: Financial transmission mecha
nism
   - Address this evaluator finding explicitly: Specific financial figures o
r details on the impact of events
   - How has market performance affected AAPL's current risk picture?
   - What is AAPL's current financial condition?
   - What SEC evidence is available for AAPL?
   - How have recent news events impacted AAPL's risk profile?
   - How does this run satisfy the saved success check: Include supporting a
rticle evidence, a financial implication, and uncertainty; state explicitly
when no material effect is established?
   - How does this run satisfy the saved success check: The next report supp
lies relevant evidence and states remaining uncertainty for this finding?

======================================================================
END OF FINAL RESEARCH REPORT
======================================================================
```

9. Ten-year market context

The original notebook's 10-year graph is preserved as a reusable reporting function. It uses adjusted closing prices and reports the start price, end price, and total return.

In [12]:
```
try:
    price_chart = plot_10_year_stock_price(YahooFinanceClient(), TICKER)
    if price_chart is None:
        display(Markdown(f"*No 10-year price history was available for {TICK
    else:
        print(price_chart.summary)
except Exception as exc:
    display(Markdown(f"*The 10-year chart could not be loaded: {type(exc).__
```



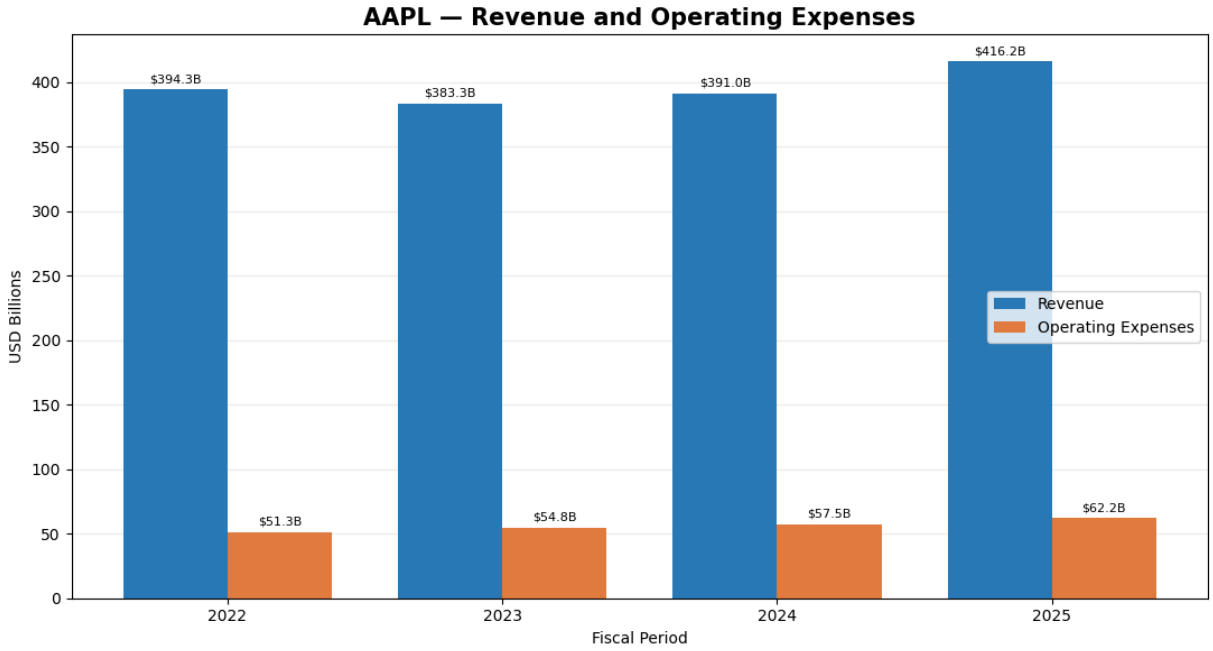

```
10-year price summary: $26.88 → $343.46 (+1,177.7%)
```

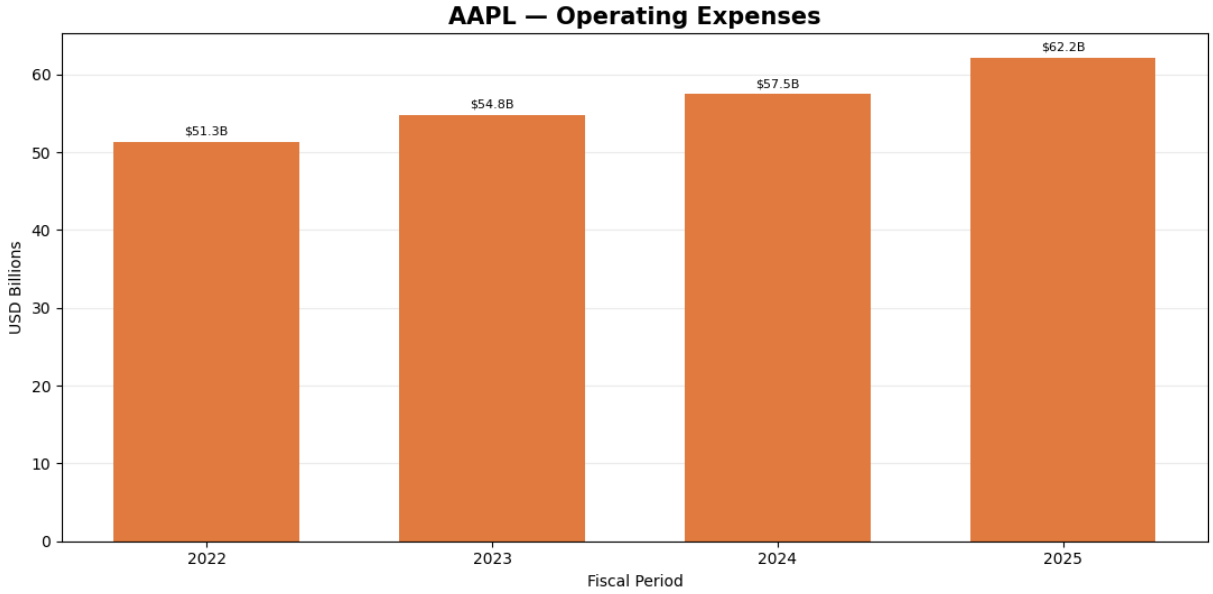
10. Revenue and operating-expense context

The financial tool retains directly reported operating expenses when available and fills only missing periods with `Gross Profit − Operating Income`. Generic total-cost rows are deliberately excluded because they may include cost of revenue. The first chart compares revenue with operating expenses across up to five available annual periods; the second isolates the operating-expense trend.

In [13]:
```
financial_observations = result.get("observations", {}).get("financials", {}
revenue_expense_figure = plot_revenue_and_operating_expenses(
    financial_observations,
    TICKER,
)
if revenue_expense_figure is None:
    display(Markdown("*Revenue data was unavailable for the comparison chart

expense_figure = plot_operating_expenses(financial_observations, TICKER)
if expense_figure is None:
    display(Markdown("*Operating-expense data was unavailable for the standa
```





11. Evaluation evidence and iteration

Each news analysis receives a **structural completeness and source-traceability score** with 25 points for each of the following:

- at least one concrete observation;
- a clear interpretation;
- an explicit uncertainty or missing-information statement; and
- at least two evidence-based observations.

A separate substantive score checks whether the observations go beyond headline/source metadata and whether the interpretation explains financial reasoning or event significance. Thus, 100/100 on the structural rubric establishes expected fields and traceability—not perfect financial analysis. A structural failure **or substantive score**

below 60 triggers revision up to `MAX_OPTIMIZER_ITERATIONS` ; unresolved work is labeled `insufficient_reasoning_or_evidence` . The complete report is separately scored for completeness, grounding, reasoning, consistency, sources, and clarity; LangGraph routes a failed draft through the report optimizer and back to evaluation up to `MAX_REPORT_REVISIONS` . Before/after scores and every graph node are retained below.

In [14]:

```
report_iteration_rows = [
    {
        "evaluation": index,
        "score": evaluation.get("score"),
        "passed": evaluation.get("passed"),
        **evaluation.get("criteria", {}),
    }
    for index, evaluation in enumerate(result.get("report_evaluation_history
]
display(Markdown("### Complete-report evaluator-optimizer history"))
display(pd.DataFrame(report_iteration_rows))
print(json.dumps({
    "report_revisions": result.get("report_revision_history", []),
    "graph_trace": result.get("graph_trace", []),
}, indent=2, default=str))

iteration_rows = []
for index, article in enumerate(news_analysis.get("articles", []), start=1):
    for revision, evaluation in enumerate(article.get("evaluation_history",
        iteration_rows.append({
            "article": index,
            "revision": revision,
            "structural_score": evaluation.get("score"),
            "structural_passed": evaluation.get("passed"),
            "substantive_reasoning_score": evaluation.get("substantive_reaso
            "feedback": "; ".join(evaluation.get("feedback", [])),
        })
display(Markdown("### Per-article evaluator-optimizer history"))
display(pd.DataFrame(iteration_rows))

request_router = ResearchRequestRouter()
routing_examples = [
    ("market-only routing", "Analyze price, volatility, and volume", "market
    ("financial-only routing", "Review revenue, debt, and the balance sheet"
    ("news-only routing", "Summarize recent news sentiment and catalysts", "
    ("comprehensive routing", "Compare market returns with earnings and rece
]
behavior_rows = []
for requirement, request_text, expected_route in routing_examples:
    route_result = request_router.route(request_text, TOOLS.keys(), has_news
    behavior_rows.append({"requirement": requirement, "expected": expected_r

article_router = SpecialistRouter()
routing_fixtures = [
    ("earnings-release fixture", {"title": "Example reports revenue and earn
    ("market-event fixture", {"title": "Example shares fall after downgrade"
    ("general-news fixture", {"title": "Example appoints a director", "raw_t
```

```
]
for requirement, article_fixture, expected_specialist in routing_fixtures:
    actual_specialist = article_router.route(article_fixture)["specialist"]
    behavior_rows.append({"requirement": requirement, "expected": expected_s

tool_choice_fixtures = [
    ("price-volatility tool choice", "Analyze price volatility", "market", {
    ("liquidity tool choice", "Assess liquidity and debt", "financial", {"ob
]
for requirement, request_text, route_name, planner_response, expected_tools
    fixture_planner = PlannerAgent(lambda _prompt, response=planner_response
    fixture_plan = fixture_planner.plan(TICKER, TOOLS.keys(), request_text,
    behavior_rows.append({"requirement": requirement, "expected": ", ".join(

report_evaluator = EvaluatorAgent(lambda _prompt: "{}")
weak_report_result = report_evaluator.evaluate_report("# Note\n\nLooks fine.
strong_report_result = result.get("report_evaluation", {})
behavior_rows.extend([
    {"requirement": "weak complete-report test", "expected": "fail", "actual
    {"requirement": "strong complete-report test", "expected": "pass", "actu
])
saved_lessons = first_result.get("memory_entry", {}).get("lessons", [])
applied_lessons = second_result.get("plan", {}).get("applied_memory_actions"
saved_actions = {lesson.get("next_run_action") for lesson in saved_lessons}
planned_actions = {lesson.get("next_run_action") for lesson in applied_lesso
executed_actions = set(second_result.get("news_analysis", {}).get("planning_
memory_behavior_passed = bool(saved_actions) and saved_actions.issubset(plan
behavior_rows.append({"requirement": "two-run actionable-memory test", "expe
display(Markdown("### Requirement-specific behavioral demonstration"))
display(pd.DataFrame(behavior_rows))

test_run = subprocess.run(
    [sys.executable, "-m", "pytest", "-q"],
    cwd=PROJECT_ROOT,
    capture_output=True,
    text=True,
    check=False,
)
display(Markdown("### Behavioral and regression test results"))
print(test_run.stdout.strip())
if test_run.returncode:
    print(test_run.stderr.strip())
print(f"pytest exit code: {test_run.returncode}")
```

Complete-report evaluator–optimizer history

	evaluation	score	passed	completeness	grounding	reasoning	consistency	source
0	0	81.67	False	100.0	100.0	30.0	100.0	60.
1	1	89.17	True	100.0	100.0	75.0	100.0	60.

```
{
  "report_revisions": [
    {
      "iteration": 1,
      "before_score": 81.67,
      "after_score": 89.17,
      "feedback": [
        "Improve reasoning; current score is 30/100.",
        "Improve sources; current score is 60/100.",
        "No report can pass while these criteria are below 60/100: reasonin
g."
      ]
    }
  ],
  "graph_trace": [
    "plan_request",
    "comprehensive_research",
    "validate_observations",
    "reflect",
    "synthesize",
    "evaluate_report",
    "revise_report",
    "evaluate_report",
    "finalize"
  ]
}
```


Per-article evaluator–optimizer history

	article	revision	structural_score	structural_passed	substantive_reasoning_score
0	1	0	100	True	50
1	1	1	100	True	50
2	1	2	100	True	50
3	1	3	100	True	50
4	2	0	100	True	100
5	3	0	50	False	50
6	3	1	100	True	0
7	3	2	100	True	0
8	3	3	100	True	0
9	4	0	100	True	50
10	4	1	100	True	50
11	4	2	100	True	50
12	4	3	100	True	50

	article	revision	structural_score	structural_passed	substantive_reasoning_score
13	5	0	100	True	100
14	6	0	100	True	50
15	6	1	100	True	50
16	6	2	100	True	50
17	6	3	100	True	50
18	7	0	100	True	0
19	7	1	100	True	0
20	7	2	100	True	0
21	7	3	100	True	0
22	8	0	75	False	0
23	8	1	100	True	50
24	8	2	100	True	50

	article	revision	structural_score	structural_passed	substantive_reasoning_score
25	8	3	100	True	50

Requirement-specific behavioral demonstration

	requirement	expected	actual	passed
0	market-only routing	market	market	True
1	financial-only routing	financial	financial	True
2	news-only routing	news	news	True
3	comprehensive routing	comprehensive	comprehensive	True
4	earnings-release fixture	earnings	earnings	True
5	market-event fixture	market	market	True
6	general-news fixture	news	news	True
7	price-volatility tool choice	price_data	price_data	True
8	liquidity tool choice	balance_sheet, cash_flow	balance_sheet, cash_flow	True
9	weak complete-report test	fail	fail	True
10	strong complete-report test	pass	pass	True
11	two-run actionable-memory test	lesson persisted → action planned → action exe...	lessons=4; planned=3; news_actions_executed=3;...	True

Behavioral and regression test results

```
.............................................................. [ 59%]
............s.............................  [100%]
============================== warnings summary ===========================
====
../../../../.local/share/virtualenvs/brandonwirgau-gZAqjFTS/lib/python3.13/s
ite-packages/langgraph/cache/base/__init__.py:8
  /Users/brandonwirgau/.local/share/virtualenvs/brandonwirgau-gZAqjFTS/lib/p
ython3.13/site-packages/langgraph/cache/base/__init__.py:8: LangChainPending
DeprecationWarning: The default value of `allowed_objects` will change in a
future version. Pass an explicit value (e.g., allowed_objects='messages' or
allowed_objects='core') to suppress this warning.
    from langgraph.checkpoint.serde.jsonplus import JsonPlusSerializer

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
111 passed, 1 skipped, 1 warning, 31 subtests passed in 4.99s
pytest exit code: 0
```

12. Memory and learning across runs

Run 1 stores structured `weakness → next_run_action → success_check` lessons in an isolated memory file. Run 2 receives those lessons, adds each action to its plan, and reanalyzes the exact same collected observations and source articles. The comparison below checks persistence, retrieval, changed planning, action execution, output evidence, and the initial substantive-reasoning score. Different question wording alone is never counted as learning; if the targeted quality measure does not improve, the notebook says so explicitly.

In [15]:
```python
first_plan = first_result.get("plan", {})
second_plan = second_result.get("plan", {})
memory_context = second_plan.get("memory_context", [])
saved_lessons = first_result.get("memory_entry", {}).get("lessons", [])
applied_lessons = second_plan.get("applied_memory_actions", [])
target_lesson = next((lesson for lesson in saved_lessons if "financial" in l
target_action = target_lesson.get("next_run_action", "")
plan_applied = target_lesson in applied_lessons
news_actions = second_result.get("news_analysis", {}).get("planning_actions"
action_executed = target_action in news_actions or target_action in second_r
same_source_articles = first_result.get("news_analysis", {}).get("source_art
same_tool_observations = all(first_result.get("observations", {}).get(name)

def initial_reasoning_scores(run_result):
    return [article.get("evaluation_history", [{}])[0].get("substantive_reas

first_initial_scores = initial_reasoning_scores(first_result)
second_initial_scores = initial_reasoning_scores(second_result)
first_initial_average = sum(first_initial_scores) / len(first_initial_scores
second_initial_average = sum(second_initial_scores) / len(second_initial_sco
quality_improved = first_initial_average is not None and second_initial_aver
supporting_excerpts = [article.get("optimized_analysis", {}).get("interpreta
output_addresses_action = bool(supporting_excerpts) and (not target_action o
memory_rows = [
    {"evidence": "Saved weakness", "run_1": target_lesson.get("weakness", "N
```

```
    {"evidence": "Planning requirement", "run_1": "Absent before feedback",
    {"evidence": "Evidence snapshot", "run_1": "Collected from providers", "
    {"evidence": "Action execution", "run_1": "Not available", "run_2": "App
    {"evidence": "Initial substantive-reasoning average", "run_1": first_ini
    {"evidence": "Targeted output behavior", "run_1": "Feedback identified a
]
display(pd.DataFrame(memory_rows))
display(Markdown("### Required versus optional tool decisions"))
display(pd.DataFrame([
    {"run": "Run 1", "required_tools": ", ".join(first_plan.get("required_to
    {"run": "Run 2", "required_tools": ", ".join(second_plan.get("required_t
]))
print("Saved Run 1 feedback and structured lessons:")
print(json.dumps(first_result.get("memory_entry", {}), indent=2, default=str
print("Representative Run 2 output excerpts:")
print(json.dumps(supporting_excerpts[:3], indent=2, default=str))
if plan_applied and action_executed and quality_improved and output_addresse
    print("Learning interpretation: demonstrated substantive cross-run impro
elif plan_applied and action_executed:
    print("Learning interpretation: memory retrieval and action execution su
else:
    print("Learning interpretation: no complete feedback → action → output c
```

	evidence	run_1	run_2	passed
0	Saved weakness	Financial transmission mechanism	Retrieved from memory	True
1	Planning requirement	Absent before feedback	Address this evaluator finding explicitly: Fin...	True
2	Evidence snapshot	Collected from providers	Exact source articles and tool observations re...	True
3	Action execution	Not available	Applied to planning, news analysis, and report...	True
4	Initial substantive-reasoning average	37.5	50.0	True
5	Targeted output behavior	Feedback identified a weakness	The article provides evidence of market activi...	True

Required versus optional tool decisions

	run	required_tools	optional_choices
0	Run 1	price_data, sec_edgar, financials	None
1	Run 2	price_data, sec_edgar, financials	None

Saved Run 1 feedback and structured lessons:

```
{
  "date": "2026-10-18T03:10:02.449603+00:00",
  "symbol": "AAPL",
  "summary": "Here's a summary of reusable research lessons from the AAPL ru
n in 120 words:\n\nEvidence coverage is essential for a comprehensive risk p
icture. However, data quality issues arise due to missing financial figures
and uncertain event significance. To improve, researchers should verify fina
ncial transmission mechanisms and cite supporting article evidence. Clarity
and consistency are also crucial for a well-structured report. The verificat
ion scope should explicitly compare filing periods and provider statement pe
riods. The use of deterministic observations for verified financial metrics
can help. The interpretation should consider both historical return and meas
ured price risk. Finally, researchers should verify detected events and ackn
owledge uncertainty in financial significance. These lessons can inform futu
re research and improve the quality of AAPL investment research reports.",
  "weaknesses": [
    "Substantive financial reasoning and event significance"
  ],
  "lessons": [
    {
      "weakness": "News analysis lacks financial implications",
      "next_run_action": "Explain how each material event could affect reven
ue, costs, cash flow, or valuation",
      "success_check": "Include supporting article evidence, a financial imp
lication, and uncertainty; state explicitly when no material effect is estab
lished"
    },
    {
      "weakness": "Substantive financial reasoning and event significance",
      "next_run_action": "Explain how each material event could affect reven
ue, costs, cash flow, or valuation",
      "success_check": "Include supporting article evidence, a financial imp
lication, and uncertainty; state explicitly when no material effect is estab
lished"
    },
    {
      "weakness": "Financial transmission mechanism",
      "next_run_action": "Address this evaluator finding explicitly: Financi
al transmission mechanism",
      "success_check": "The next report supplies relevant evidence and state
s remaining uncertainty for this finding"
    },
    {
      "weakness": "Specific financial figures or details on the impact of ev
ents",
      "next_run_action": "Address this evaluator finding explicitly: Specifi
c financial figures or details on the impact of events",
      "success_check": "The next report supplies relevant evidence and state
s remaining uncertainty for this finding"
    }
  ]
}
```

Representative Run 2 output excerpts:

```
[
  "The article provides evidence of market activity and stake increase, but
```

```
lacks specific financial details to assess the impact on revenue, costs, cas
h flow, or valuation.",
  "The supplied article evidence does not by itself establish how the report
ed event affects revenue, costs, cash flow, or valuation; a material financi
al effect therefore remains uncertain.",
  "The article provides evidence relevant to share price, market activity, a
nalyst commentary, and valuation."
]
Learning interpretation: demonstrated substantive cross-run improvement. Run
1 feedback produced a structured action; Run 2 applied it to identical evide
nce and improved the targeted initial reasoning score with visible output su
pport.
```

13. Limitations, responsible use, and conclusion

- Yahoo Finance, NewsAPI, Google News RSS, and SEC EDGAR can be delayed, unavailable, rate-limited, incomplete, or inconsistent.
- News summaries may contain only provider snippets; source links should be reviewed before decisions.
- Keyword extraction and routing are intentionally transparent but cannot capture every nuance.
- Local-model output is nondeterministic and may still omit context; deterministic validation reduces but cannot eliminate that risk.
- Google News RSS is a coverage fallback, not a replacement for primary or official evidence.
- FRED and Alpha Vantage remain scaffolds and are not represented as report evidence.
- Historical performance, evaluator scores, and model interpretations do not predict future returns.

The final system preserves the original financial-research workflow while incorporating the updated notebook's operating-expense analysis, financial trend charts, and resilient multi-source news retrieval. Its modular adapters, explicit state, provenance, bounded revision loop, deterministic report, persistent memory, offline tests, and generated diagrams make the work reproducible and auditable.

Appendix — companion audit artifact

The complete workflow state is written to a companion HTML file rather than repeated in the main report. This preserves every plan, observation, article, evaluator result, revision, graph node, and memory artifact while keeping the review copy compact.

In [16]:
```
audit_path = PROJECT_ROOT / "notebooks" / "multi_agent_financial_analysis_au
audit_json = json.dumps(result, indent=2, default=str)
audit_document = f"""<!doctype html>
<html lang="en"><head><meta charset="utf-8"><title>Multi-Agent Financial Ana
```

```
<style>body{{font-family:system-ui,sans-serif;max-width:1200px;margin:2rem a
</head><body><h1>Multi-Agent Financial Analysis — Complete Audit State</h1>
<p>Ticker: {html.escape(str(result.get('symbol', '')))}. Generated by the ex
<pre>{html.escape(audit_json)}</pre></body></html>"""
audit_path.write_text(audit_document, encoding="utf-8")
audit_summary = [{"artifact": key, "record_count": len(value) if isinstance(
display(pd.DataFrame(audit_summary))
print(f"Complete audit artifact: {audit_path}")
print(f"Audit size: {audit_path.stat().st_size:,} bytes")
```

	artifact	record_count
0	symbol	1
1	research_request	1
2	remember	1
3	request_route	1
4	selected_specialists	3
5	plan	11
6	observations	4
7	specialist_analyses	3
8	news_analysis	21
9	validation	0
10	reflection	6
11	report	1
12	draft_report	1
13	best_report	1
14	best_report_evaluation	7
15	report_validation	0
16	report_evaluation	7
17	report_evaluation_history	2
18	report_revision_history	1
19	report_revision_count	1
20	max_report_revisions	1
21	graph_trace	9
22	evidence_snapshot	25
23	evidence_reused	1

```
Complete audit artifact: /Users/brandonwirgau/Projects/AAI/520/Final Projec
t/notebooks/multi_agent_financial_analysis_audit.html
Audit size: 569,567 bytes
```